



SCHOOL LEVEL

Class: 11 PCM

Date:16/11/2024

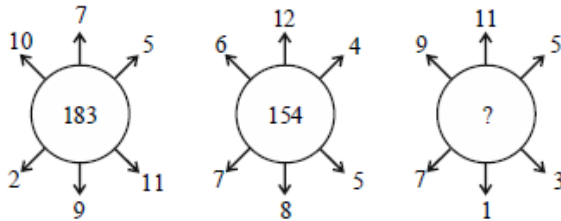
Marks: 100

Name of the Student: _____

GENERAL INSTRUCTIONS

1. Read the questions carefully.
2. Attempt all the questions.
3. No negative marking.
4. Answers to be marked in the OMR sheet. Darken the bubble completely. Partially darkened bubbles will not be considered.
5. Question numbers 1 to 100 carries 1mark each.

1. Find the missing character



a. 86

b. 73

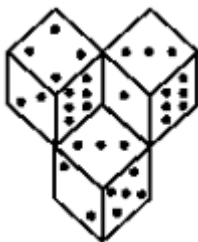
c. 33

d. 104

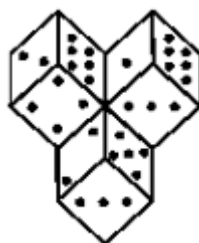
2 Find the water image of X



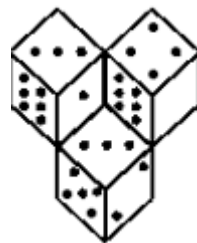
(X)



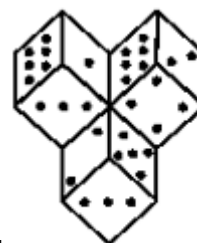
a.



b.

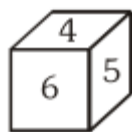
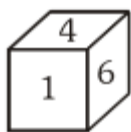


c.



d.

3 Two positions of a dice are shown below. When number '1' is on the top, what number will be at the bottom?



a. 3

b. 5

c. 2

d. 6

4. CDEF : MNOP :: UVWX : ?

A. DEFG

B. EFGH.

C. GHIJ

D. ABCD

5. Complete the analogy:

Cardiologist : Heart :: Chiropractor :

A. Ear

B. Joint

C. Brain

D. Skull

6. Guess the next number: 19 : 361 :: 66 : ____

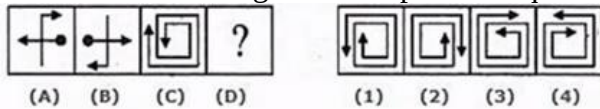
A. 693

B. 256

C. 4356.

D. 5346

7. Which suitable figure will replace the question mark based on the figures given in question?



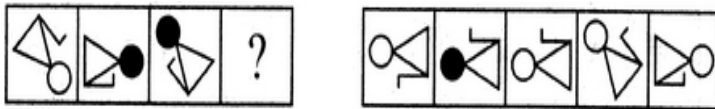
A. 1

B. 2.

C. 3

D. 4

8. Based on the figure in the question find a suitable figure from the five figures in the option



A. 1

B. 2.

C. 3

D. 4

9. In a row all the persons are facing north, Rahul is 33rd from the left end and in the right side of Rahul, there are only 16 persons. Find out total number of persons in this queue?

A. 49

B. 50

C. 51

D. None of these

10. Five friends are standing in a line. Nishu is taller than Riya but shorter than Pooja. Amrita is the shortest. Riya is shorter than Nishu but taller than Nikita. Who is the second tallest?

A. Amrita.

B. Pooja.

C. Riya

D. Nishu

11. There are 25 boys in a horizontal row. Rahul was shifted by three places towards his right side and he occupies the middle position in the row. What was his Original position from the left end of the row

A. 15.

B. 14.

C. 12.

D. 10

12. Find the odd one out: 396, 462, 572, 427, 671, 264

A. 396.

B. 427

C. 671

D. 264

13. On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec, 2006 ?

A. Sunday

B. Thursday.

C. Tuesday

D. Friday

14. Clock is started at noon. By 10 minutes past 5, the hour hand has turned through:

A. 145°.

B. 150°.

C. 155°

D. 160°

15. The average of 20 numbers is zero. Of them, at the most, how many may be greater than zero?

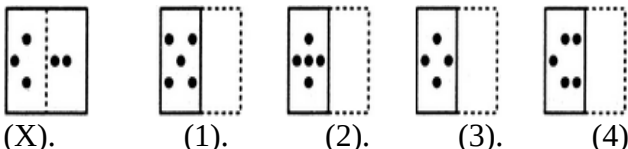
A. 0.

B. 1.

C. 10.

D. 19

16. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.



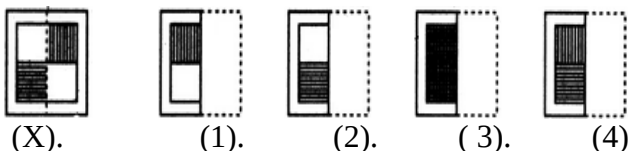
A. 1.

B. 2.

C. 3

D. 4

17. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.



A. 1.

B. 2.

C. 3.

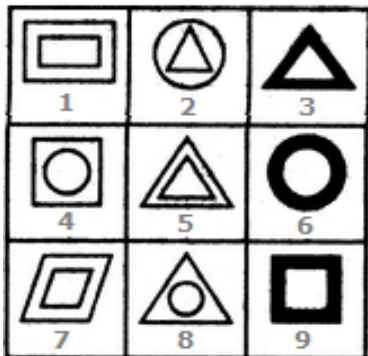
D. 4

18. Find the next term in the alphanumeric series: Z1A, X2D, V6G, T21J, R88M, P445P, ?

- A. N2676P. B. N2676S. C. T2670N. D. T2676S

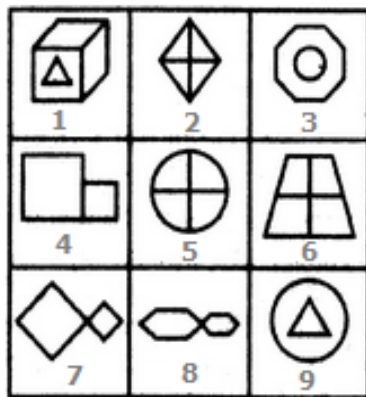
Group the given figures in Qn. 19 & 20 into three classes using each figure only once.

19.



- A) 1,5,7 ; 2,4,6 ; 3,9,8 B) 1,5,7 ; 2,4,8 ; 3,6,9.
C) 1,4,7 ; 2,5,8 ; 3,6,9 D) 1,7,9 ; 3,5,8 ; 2,4,6

20.



- A) 1,3,9 ; 2,5,6 ; 4,7,8. B) 1,3,9 ; 2,7,8 ; 4,5,6
C) 1,2,4 ; 3,5,7 ; 6,8,9 D) 1,3,6 ; 2,4,8 ; 5,7,9

21. Peter is in the East of Tom, and Tom is in the North of John. Mike is in the South of John, then in which direction of Peter is Mike?

- A) South-East B) South-West C) South. D) North-East

22. There are four pipes used to fill a tank at a constant rate. Altogether, the four pipes are able to fill up the tank in 4 hours. The owner wishes to reduce this time taken to 2 hours. How many more pipes filling up the tank at a constant rate will they need to install?

- A) 2. B) 8. C) 4. D) 6

23. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?

- A) 30% B) 70%. C) 100%. D) 250%

24. A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?

- A) 3. B) 4. C) 5. D) 6

25. Four defensive football players are chasing the opposing wide receiver, who has the ball. Calvin is directly behind the ball carrier. Jenkins and Burton are side by side behind Calvin. Zeller is behind Jenkins and Burton. Calvin tries for the tackle but misses and falls. Burton trips. Which defensive player tackles the receiver?

- A) Burton. B) Zeller C) Jenkins D) Calvin

26. Select the correct meaning of the word "obfuscate":

- a) Simplify b) Confuse c) Clarify d) Support

27. Which of the following is a synonym of “pernicious”?
- a) Harmless b) Beneficial c) Destructive d) Friendly
28. Choose the correct antonym of “capricious”:
- a) Unpredictable b) Constant c) Erratic d) Arbitrary
29. Which word best fits the sentence?
- _The _____ politician swayed the audience with his manipulative speech._
- a) candid b) disingenuous c) amiable d) effusive
30. What does the term “belligerent” mean?
- a) Friendly b) Hostile c) Indifferent d) Relaxed
31. Identify the correct noun form in the sentence:
- _The President’s _____ was clear and concise._
- a) announce b) announcer c) announcement d) announcing
32. Which sentence uses a demonstrative pronoun correctly?
- a) That are my books. c) These is the way.
b) Those are your options. d) This are the result.
33. Complete the sentence using the correct verb form:
- _By next year, they _____ the project._
- a) will complete c) will have completed
b) will be completing d) have completed
34. Choose the correct adverb for the sentence:
- _She handled the situation _____._
- a) with care b) carefully c) carelessly d) careful
35. Select the correct comparative adjective:
- _This test is _____ than the previous one._
- a) more easier b) easier c) easy d) easiest
36. Choose the correct preposition:
- _The decision depends _____ the outcome of the meeting._
- a) from b) on c) in d) for
37. Complete the sentence with the correct conjunction:
- She studied hard _____ she could pass the exam.
- a) so that b) and c) but d) while
38. Select the correct modal verb:
- He _____ take an umbrella since it’s going to rain.
- a) should b) must c) can d) might
39. Which sentence correctly uses a gerund?
- a) He is fond of to play football. c) He is interested to play football.
b) He enjoys playing football. d) He enjoys to play football.
40. Choose the correct participle:
- The _____ tree provided much-needed shade.
- a) fallen b) falling c) fall d) fell
41. Which of the following sentences is in the passive voice?
- a) The teacher explains the lesson. c) The teacher will explain the lesson.
b) The lesson is explained by the teacher. d) The teacher is explaining the lesson.
42. Choose the correct indirect speech:
- He said, “I have completed the work.”
- a) He said that he had completed the work. c) He said that he has completed the work.
b) He says he completed the work. d) He said that he will complete the work.
43. Form a question from the given sentence:
- _They are going to the party._
- a) Are they going to the party? c) They are going to the party, right?
b) Will they be going to the party? d) Is they going to the party?

44. Identify the type of sentence:

Please submit the report by tomorrow.

a) Declarative

c)

d)

b) Imperative

Interrogative

Exclamatory

Clauses, Idioms, and Proverbs

45. Identify the subordinate clause in the sentence:

If you study hard, you will succeed.

a) If you study hard

c) Study hard

b) You will succeed

d) Succeed

46. What does the idiom "bite the bullet" mean?

a) To avoid responsibility

c) To procrastinate

b) To face a difficult situation with courage

d) To react impulsively

47. Choose the correct meaning of the proverb "The early bird catches the worm":

a) Early risers get the best opportunities.

c) Preparation is the key to success.

b) Hard work pays off in the end.

d) Opportunities are rare

48. Which of the following sentences is punctuated correctly?

a) However, I still don't agree with you.

c) However I still, don't agree with you.

b) However I still don't agree with you.

d) However; I still don't agree with you.

49. What is the purpose of a colon in the following sentence?

She brought three items: a pen, a notebook, and a calculator.

a) To introduce a list

c) To show possession

b) To separate two independent clauses

d) To express emphasis

50. Choose the correct transformation of this sentence:

She is too tired to work.

a) She is so tired that she cannot work.

c) She is tired but can work.

b) She is very tired, so she can work.

d) She is tired yet working

51. The number of students who take both the subjects Mathematics and Chemistry is 30. This represents 10% of the enrolment in Mathematics and 12% of the enrolment in Chemistry.

How many students take at least one of these two subjects?

(a) 520

(b) 490

(c) 560

(d) 480

52. The domain of the function $f(x) = \log_2[\log_3(\log_4 x)]$ is

(a) $(-\infty, 4)$

(b) $(4, \infty)$

(c) $(0, 4)$

(d) $(1, \infty)$

53. The value of x in $\left(0, \frac{\pi}{2}\right)$ satisfying the equation $\sin x \cos x = \frac{1}{4}$ is

- (a) $\frac{\pi}{6}$
- (b) $\frac{\pi}{3}$
- (c) $\frac{\pi}{8}$
- (d) $\frac{\pi}{12}$

54. If $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, $\sin \alpha = \frac{4}{5}$ and $\cos(\alpha + \beta) = -\frac{12}{13}$, then $\sin \beta$ is equal to

(a) $\frac{63}{65}$

(b) $\frac{61}{65}$

(c) $\frac{3}{5}$

(d) $\frac{5}{13}$

55. If $a_1 = 1$ and $a_n = na_{n-1}$ for all positive integers $n \geq 2$, then a_5 is equal to

(a) 125

(b) 120

(c) 100

(d) 24

56. If $(\sqrt{5} + \sqrt{3}i)^{33} = 2^{49}z$, then modulus of complex number z is

(a) 1

(b) $\sqrt{2}$

(c) $2\sqrt{2}$

(d) 4

57. The equation whose roots are the squares of the roots of the equation $2x^2 + 3x + 1 = 0$, is

(a) $4x^2 + 5x + 1 = 0$

(b) $4x^2 - x + 1 = 0$

(c) $4x^2 - 5x - 1 = 0$

(d) $4x^2 - 5x + 1 = 0$

58. If x satisfies the inequations $2x - 7 < 11$ and $3x + 4 < -5$, then x lies in the interval

(a) $(-\infty, 3)$

(b) $(-\infty, 2)$

(c) $(-\infty, -3)$

(d) $(-\infty, \infty)$

59. All the words that can be formed using alphabets A, H, L, U, R are written as in a dictionary (no alphabet is repeated). Then, the rank of the word 'RAHUL' is

(a) 71

(b) 72

(c) 73

(d) 74

60. Let $[x]$ denotes the greatest integer less than or equal to x . If $x = (\sqrt{3} + 1)^5$, then $[x]$ is equal to
- (a) 75
 - (b) 50
 - (c) 152
 - (d) 76
61. The sum of all two digit natural numbers which leave a remainder 5 when they are divided by 7 is equal to
- (a) 702
 - (b) 715
 - (c) 589
 - (d) 602
62. The ratio by which the line $2x + 5y - 7 = 0$ divides the straight line joining the points $(-4, 7)$ and $(6, -5)$ is
- (a) 1 : 4
 - (b) 1 : 2
 - (c) 1 : 1
 - (d) 2 : 3

63. If the lines $y = 3x + 1$ and $2y = x + 3$ are equally inclined to the line $y = mx + 4$, $\left(\frac{1}{2} < m < 3\right)$, then the values of m is
- (a) $\frac{1}{7}(1 \pm 5\sqrt{3})$
 (b) $\frac{1}{7}(1 \pm 5\sqrt{5})$
 (c) $\frac{1}{7}(1 \pm 5\sqrt{2})$
 (d) $\frac{1}{7}(1 \pm 2\sqrt{35})$
64. The slope of the straight line joining the centre of the circle $x^2 + y^2 - 8x + 2y = 0$ and the vertex of the parabola $y = x^2 - 4x + 10$ is
- (a) $\frac{-5}{2}$
 (b) $\frac{-7}{2}$
 (c) $\frac{-3}{2}$
 (d) $\frac{7}{2}$
65. The distance between the X-axis and the point $(3, 12, 5)$ is
- (a) 3
 (b) 13
 (c) 15
 (d) 5
66. $\lim_{x \rightarrow 0} \frac{\log(1+3x^2)}{x(e^{5x}-1)}$ is equal to
- (a) $\frac{3}{5}$
 (b) $\frac{5}{3}$
 (c) $\frac{-3}{5}$
 (d) $\frac{-5}{3}$
67. If $y = f(x^2 + 2)$ and $f'(3) = 5$, then $\frac{dy}{dx}$ at $x = 1$ is
- (a) 5
 (b) 25
 (c) 20
 (d) 10
68. The truth values of p, q and r for which $(p \wedge q) \vee (\sim r)$ has truth value F are respectively
- (a) F, T, F
 (b) T, F, F
 (c) F, F, T
 (d) F, F, F

69. The standard deviation of 9, 16, 23, 30, 37, 44, 51 is
 (a) 7
 (b) 9
 (c) 12
 (d) 14
70. If two dice are thrown simultaneously, then the probability that the sum of the numbers which come up on the dice to be more than 5 is
 (a) $\frac{5}{36}$
 (b) $\frac{1}{6}$
 (c) $\frac{5}{18}$
 (d) $\frac{13}{18}$
71. A force F is applied onto a square plate of side L . If the percentage error in determining L is 2% and that in F is 4%, the permissible percentage error in determining the pressure is
 a. 2% c. 6%
 b. 4% d. 8%
72. The radius of gyration about an axis through the center of a hollow sphere with external radius a and internal radius b is
 a. $\sqrt{\frac{2(a^3-b^3)}{5(a^5-b^5)}}$ c. $\sqrt{\frac{1(a^5-b^5)}{2(a^3-b^3)}}$
 b. $\sqrt{\frac{1(a^4-b^4)}{4(a^2-b^2)}}$ d. $\sqrt{\frac{2(a^5-b^5)}{5(a^3-b^3)}}$
73. A ball dropped from a point A falls down vertically to C , through the midpoint B . The descending time from A to B and that from A to C are in the ratio
 a. 1:1 b. $1:\sqrt{2}$ c. 1:3
 d. 1:2
74. A cricket ball is hit at an angle of 30° to the horizontal with a kinetic energy E . Its kinetic energy when it reaches the highest point is
 a. $\frac{E}{2}$ b. 0 c. $\frac{3E}{4}$ d. $\frac{2E}{3}$
75. A body hanging from a massless spring stretches it by 3 cm on earth's surface. At a place 800 km above the earth's surface, the same body will stretch the spring by (Radius of earth = 6400 km)
 a. $\left(\frac{34}{27}\right)$ cm b. $\left(\frac{64}{27}\right)$ cm c. $\left(\frac{27}{64}\right)$ cm d. $\left(\frac{27}{34}\right)$ cm
76. In a horizontal pipe of non-uniform cross-section, water flows with a velocity of 1 ms^{-1} at a point where the diameter of the pipe is 20 cm. The velocity of water (in ms^{-1}) at a point where the diameter of the pipe is 5 cm is
 a. 16 b. 24 c. 8 d. 32
77. Two equal masses hung from two massless springs of spring constants k_1 and k_2 have equal maximum velocity when executing simple harmonic motion. The ratio of their amplitudes is
 a. $\left(\frac{k_1}{k_2}\right)^{1/2}$ b. $\left(\frac{k_1}{k_2}\right)$ c. $\left(\frac{k_2}{k_1}\right)$ d. $\left(\frac{k_2}{k_1}\right)^{1/2}$
78. The harmonic mode which resonates with a closed pipe of length 22 cm, when excited by a 1875 Hz source and the number of nodes present in it respectively are (velocity of sound in air = 330 ms^{-1})

- a. 1st, 1
b. 5th, 3

- c. 3rd, 2
d. 5th, 4

79. Displacement, x (in meters), of a body of mass 1 kg as a function of time, t , on a horizontal smooth surface is given as $x = 2t^2$. The work done in the first one second by the external force is

- a. 1 J
b. 2 J
c. 8 J
d. 4 J

80. A rubber cord of density d , Young's modulus Y and length L is suspended vertically. If the cord extends by a length $0.5 L$ under its own weight, then L is

- a. $\frac{Y}{2dg}$
b. $\frac{Y}{dg}$
c. $\frac{2Y}{dg}$
d. $\frac{dg}{2Y}$

81. In a thermodynamic system, Q represents the energy transferred to or from a system by heat and W represents the energy transferred to or from a system by work.

- I. $Q > 0$ and $W = 0$
II. $Q < 0$ and $W = 0$
III. $W > 0$ and $Q = 0$
IV. $W < 0$ and $Q = 0$

Which of the above will lead to an increase in the internal energy of the system?

- a. I only
b. II only
c. I and IV only
d. II and III only

82. A uniform copper rod of 50 cm length is insulated on the sides, and has its ends exposed to ice and steam respectively. If there is a layer of water 1 mm thick at each end, the temperature gradient (in $^{\circ}\text{Cm}^{-1}$) in the bar is (Assume that the thermal conductivity of copper is $400\text{Wm}^{-1}\text{K}^{-1}$ and water is $0.4\text{Wm}^{-1}\text{K}^{-1}$)

- a. 60
b. 40
c. 50
d. 55

83. A spherically uniform planet of mass 8×10^{24} kg and of radius 6×10^6 m is orbiting around the Sun. The escape velocity for the planet is close to (Take $G = 6 \times 10^{-11} \text{ N} - \text{m}^2/\text{kg}^2$)

- a. 11.2 km/s
b. 16 km/s
c. 4 km/s
d. 12.6 km/s

84. A boy is standing on a weighing machine inside a lift. When the lift goes upwards with acceleration $\frac{g}{4}$, the machine shows the reading 50 kg . wt. When the lift goes downward with acceleration $\frac{g}{4}$, the reading of the machine in kg . wt. would be

- a. 50
b. 30
c. 45.5
d. 62.5

85. If n bullets each of mass m are fired with a velocity v per second from a machine gun, the force required to hold the gun in position is

- a. mnv
b. $\frac{mv}{n^2}$
c. $\frac{mv}{n}$
d. n^2mv

86. 4.28 g of NaOH is dissolved in water and the solution is made to 250 cc. What will be the molarity of the solution?

- (a) 0.615 mol L^{-1}
(b) 0.428 mol L^{-1}
(c) 0.99 mol L^{-1}
(d) 0.301 mol L^{-1}

87. Select the correct statements from the following:

- A. Atoms of all elements are composed of two fundamental particles.
B. The mass of electron is 9.10939×10^{-31} kg.
C. All the isotopes of a given element show same chemical properties.
D. Protons and electrons are collectively known as nucleons.
E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

Choose the correct answer from the options given below:

- (a) A and E only
(b) B, C and E only
(c) A, B and C only
(d) C, D and E only

88. The size of ionic species is correctly given in order

- (a) $\text{Cl}^{7+} > \text{Si}^{4+} > \text{Mg}^{2+} > \text{Na}^+$
 (b) $\text{Na}^+ > \text{Mg}^{2+} > \text{Si}^{4+} > \text{Cl}^{7+}$

- (c) $\text{Na}^+ > \text{Mg}^{2+} > \text{Cl}^{7+} > \text{Si}^{4+}$
 (d) $\text{Cl}^{7+} > \text{Na}^+ > \text{Mg}^{2+} > \text{Si}^{4+}$

89. In which one of the following the central atom is sp^3 hybridised?

- (a) NH_4^+
 (b) BF_3

- (c) SF_6
 (d) PCl_5

90. Which of the following pairs have identical bond order?

- (a) CN^- and NO^+
 (b) CN^- and O_2^-

- (c) CN^- and CN^+
 (d) NO^+ and O_2^-

91. In which one of the following reactions, entropy decreases?

- (a) Sodium chloride is dissolved in water
 (b) Water is heated from 303 K to 353 K
 (c) Sodium bicarbonate is decomposed to $\text{Na}_2\text{CO}_3(\text{s})$, $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$
 (d) Water crystallises into ice
 (e) Dihydrogen molecule is decomposed into hydrogen atoms

92. Under isothermal conditions, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is [Given that 1 Lbar = 100 J]

- (a) 30 J
 (b) -30 J
 (c) 5 kJ
 (d) 25 J

93. In which one of the following equilibria Δn_g value is zero?

- (a) $2\text{NOCl}(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{Cl}_2(\text{g})$
 (b) $\text{Ni}(\text{s}) + 4\text{CO}(\text{g}) \rightleftharpoons \text{Ni}(\text{CO})_4(\text{g})$

- (c) $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightleftharpoons 2\text{CO}(\text{g})$
 (d) $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \rightleftharpoons 2\text{HBr}(\text{g})$

94. If the ionic product of $\text{M}(\text{OH})_2$ is 5×10^{-10} , then the molar solubility of $\text{M}(\text{OH})_2$ in 0.1 M NaOH is

- (a) $5 \times 10^{-12} \text{M}$
 (b) $5 \times 10^{-10} \text{M}$

- (c) $5 \times 10^{-8} \text{M}$
 (d) $5 \times 10^{-16} \text{M}$

95. Which of the following is not a characteristic of equilibrium:

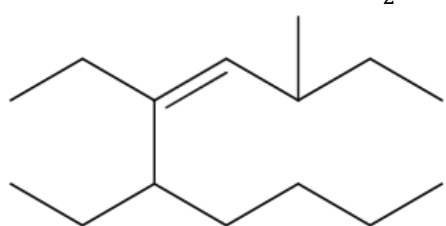
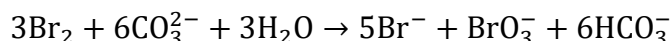
- (a) Rate is equal in both direction
 (b) Measurable quantities are constant at equilibrium
 (c) Equilibrium occurs in reversible condition
 (d) Equilibrium occurs only in open vessel at constant temperature

96. The oxidation number of iron in Fe_3O_4 is

- (a) +2
 (b) +3

- (c) +8/3
 (d) +2/3

97. In the reaction,



- (a) Bromine is oxidised and carbonate is reduced
 (b) Bromine is reduced and water is oxidised
 (c) Bromine is neither reduced nor oxidised
 (d) Bromine is both reduced and oxidised

98. The correct IUPAC name of the compound is

- (a) 3-heptyl-5-methylhept-3-ene
 (b) 5,6-diethyl-3-methyldec-4-ene
 (c) 5-butyl-3-methyloct-4-ene
 (d) 8-methyl-3-propylhex-3-ene.

99. The number of sigma (σ) and pi (π) bonds in pent-2-en-4-yne is

- (a) 13 σ bonds and no π bond
 (b) 10 σ bonds and 3 π bonds
 (c) 8 σ bonds and 5 π bonds
 (d) 11 σ bonds and 2 π bonds.

100. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following?
- (a) $-I$ effect of $-\text{CH}_3$ groups
 - (b) $+R$ effect of $-\text{CH}_3$ groups
 - (c) $-R$ effect of $-\text{CH}_3$ groups
 - (d) Hyperconjugation
